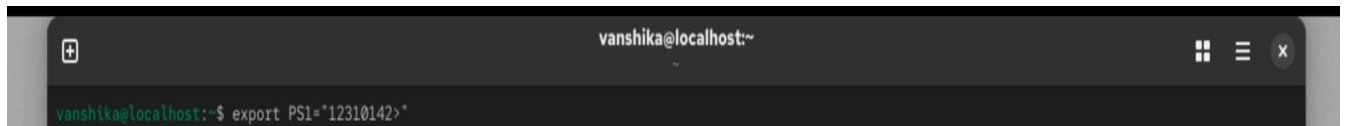


NAME: VANSHIKA PANWAR

REG NO.: 12310142

STEPS PERFORMED:

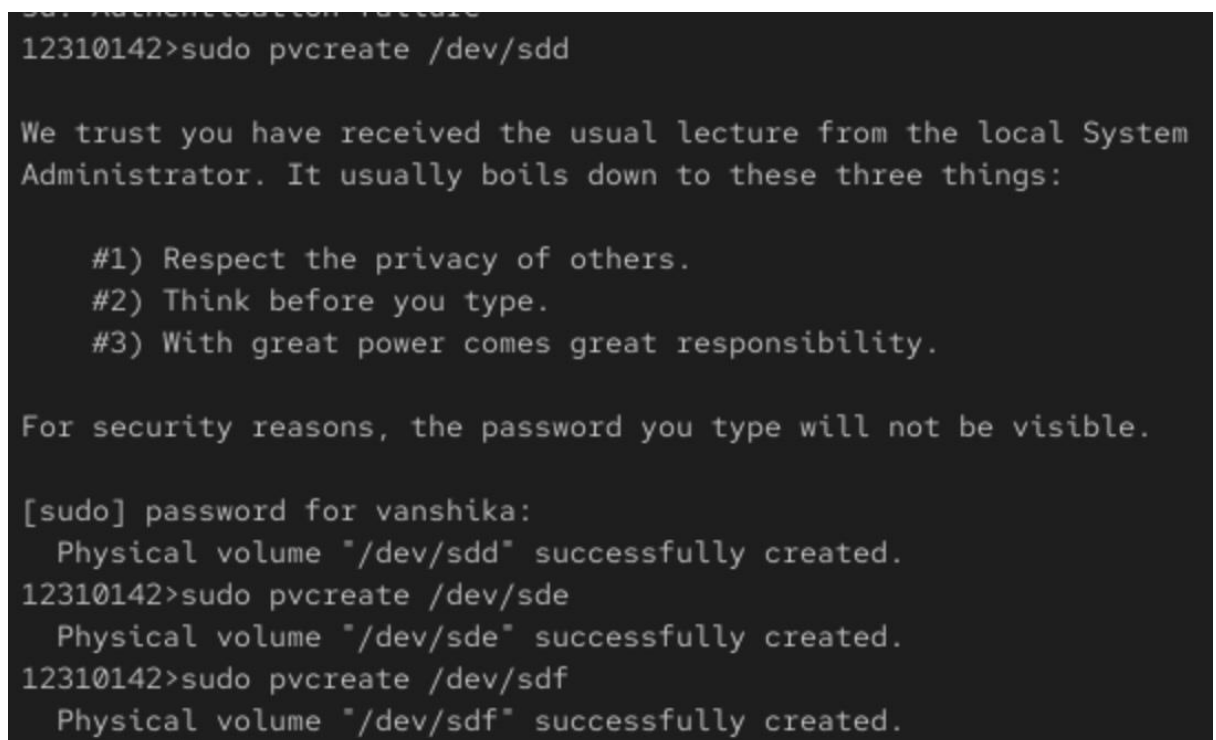
1. The shell prompt has been set to display my registration number:
12310142.

A terminal window with a dark background. The title bar shows 'vanshika@localhost:~'. The command prompt is 'vanshika@localhost:~\$' and the command entered is 'export PS1="12310142>".

```
vanshika@localhost:~$ export PS1="12310142>"
```

Shell prompt updated to: 12310142

2. Physical volume initialized

A terminal window with a dark background. The prompt is '12310142>'. The command entered is 'sudo pvcreate /dev/sdd'. The output shows a message from the local System Administrator and three rules. Then, the prompt changes to '[sudo] password for vanshika:'. After entering the password, the output shows 'Physical volume "/dev/sdd" successfully created.'. Then, the command 'sudo pvcreate /dev/sde' is entered, followed by 'Physical volume "/dev/sde" successfully created.'. Finally, the command 'sudo pvcreate /dev/sdf' is entered, followed by 'Physical volume "/dev/sdf" successfully created.'.

```
12310142>sudo pvcreate /dev/sdd

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for vanshika:
Physical volume "/dev/sdd" successfully created.
12310142>sudo pvcreate /dev/sde
Physical volume "/dev/sde" successfully created.
12310142>sudo pvcreate /dev/sdf
Physical volume "/dev/sdf" successfully created.
```

3. Volume group created:
Volume group named as proVG.

A terminal window with a dark background. The prompt is '12310142>'. The command entered is 'sudo vgcreate proVG /dev/sdd /dev/sde /dev/sdf'. The output is 'Volume group "proVG" successfully created'.

```
12310142>sudo vgcreate proVG /dev/sdd /dev/sde /dev/sdf
Volume group "proVG" successfully created
```

4. Logical Volume of 5GB created:

```
12310142>sudo lvcreate -L 5G -n proLV proVG
Logical volume "proLV" created.
```

5. Logical volume formatted and mounted:

```
12310142>sudo mkfs.ext4 /dev/proVG/proLV
mke2fs 1.47.1 (20-May-2024)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 38d65d4a-87e7-4f71-9b27-272b6a78632a
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

Created the mount point and mounted it:

```
12310142>sudo mkdir /mnt/prodata
12310142>sudo mount /dev/proVG/proLV /mnt/prodata
```

6. Configuring persistent mount for the logical volume

```
12310142>echo "/dev/proVG/proLV /mnt/prodata ext4 defaults 0 0" | sudo tee -a /etc/fstab
/dev/proVG/proLV /mnt/prodata ext4 defaults 0 0
```

7. Checked volume group details and extended logical volume to 7 GB:

```
12310142>sudo vgdisplay proVG
--- Volume group ---
VG Name                proVG
System ID
Format                 lvm2
Metadata Areas         3
Metadata Sequence No   2
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 1
Open LV                 1
Max PV                  0
Cur PV                 3
Act PV                  3
VG Size                 <7.99 GiB
PE Size                 4.00 MiB
Total PE                2045
Alloc PE / Size         1280 / 5.00 GiB
Free PE / Size           765 / <2.99 GiB
VG UUID                 62vC3f-ZtAf-m5ue-Qmqm-t2BQ-VoXL-7dTcPV

12310142>sudo lvextend -L 7G /dev/proVG/proLV
Size of logical volume proVG/proLV changed from 5.00 GiB (1280 extents) to 7.00 GiB (1792 extents).
Logical volume proVG/proLV successfully resized.
12310142>sudo resize2fs /dev/proVG/proLV
resize2fs 1.47.1 (20-May-2024)
Filesystem at /dev/proVG/proLV is mounted on /mnt/prodata; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/proVG/proLV is now 1835008 (4k) blocks long.
```

8. Verified and installed ACL package; remounted filesystem with ACL support:

```
12310142>sudo yum install acl
Updating Subscription Management repositories.
Last metadata expiration check: 0:46:44 ago on Sun 06 Jul 2025 05:46:24 PM IST.
Package acl-2.3.2-4.el10.aarch64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
12310142>sudo mount -o remount,acl /mnt/prodata
```

9. Created new user, set password, and configured ACL permissions:

```
12310142>sudo useradd prouser
12310142>sudo passwd prouser
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
12310142>sudo setfacl -m u:prouser:rwx /mnt/prodata
```

- 10.Verified the ACL Permission:

```
12310142>getfacl /mnt/prodata
getfacl: Removing leading '/' from absolute path names
# file: mnt/prodata
# owner: root
# group: root
user::rwx
user:prouser:rwx
group::r-x
mask::rwx
other::r-x
```

11. Configured ACL to restrict access for other users and extended logical volume to 7 GB:

```
12310142>sudo setfacl -m o:--- /mnt/prodata
12310142>sudo lvextend -L 7G /dev/proVG/proLV
  New size (1792 extents) matches existing size (1792 extents).
  No size change.
12310142>sudo resize2fs /dev/proVG/proLV
resize2fs 1.47.1 (20-May-2024)
The filesystem is already 1835008 (4k) blocks long.  Nothing to do!
```

VERIFICATION:

1. Verified Physical Volume Creation:

```
12310142>sudo pvs
PV          VG          Fmt  Attr  PSize    PFree
/dev/nvme0n1p3 rhel      lvm2  a--   18.41g    0
/dev/sda      project_vg lvm2  a--   1020.00m  1012.00m
/dev/sdb      project_vg lvm2  a--    <3.00g    0
/dev/sdc      project_vg lvm2  a--    <4.00g    0
/dev/sdd      proVG      lvm2  a--   1020.00m  1012.00m
/dev/sde      proVG      lvm2  a--    <3.00g    0
/dev/sdf      proVG      lvm2  a--    <4.00g    0
```

2. Verified Volume Group:

```
12310142>sudo vgs
VG          #PV #LV #SN Attr   VSize  VFree
proVG       3  1  0 wz--n- <7.99g 1012.00m
project_vg  3  1  0 wz--n- <7.99g 1012.00m
rhel        1  2  0 wz--n- 18.41g    0
```

3. Verified Logical Volume:

```
12310142>sudo lvs
LV          VG          Attr      LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
proLV       proVG       -wi-ao---- 7.00g
projectlv   project_vg  -wi-ao---- 7.00g
root        rhel        -wi-ao---- 16.41g
swap        rhel        -wi-ao---- 2.00g
```

4. Verified active mount status and confirmed final size as 7 GB after resizing

```
12310142>df -h /mnt/prodata
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/proVG-proLV   6.9G   24K  6.5G   1% /mnt/prodata
```

5. Verified projectuser has appropriate permissions; access restricted for others:

```
12310142>getfacl /mnt/prodata
getfacl: Removing leading '/' from absolute path names
# file: mnt/prodata
# owner: root
# group: root
user::rwx
user:prouser:rwx
group::r-x
mask::rwx
other:---
```

6. Verified logical volume existence, size update, and compliance with specifications:

```
12310142>sudo lvdisplay /dev/proVG/proLV
--- Logical volume ---
LV Path                /dev/proVG/proLV
LV Name                 proLV
VG Name                 proVG
LV UUID                 xdpQB2-nlvB-TQlB-wKBG-fD9M-1PUO-2yowRE
LV Write Access         read/write
LV Creation host, time  localhost.localdomain, 2025-07-06 18:24:21 +0530
LV Status                available
# open                  1
LV Size                 7.00 GiB
Current LE              1792
Segments                3
Allocation              inherit
Read ahead sectors      auto
- currently set to     256
Block device            253:3
```

HISTORY

```
2 export PS1="12310142>"
3 lsblk
4 su -
5 sudo pvcreate /dev/sdd
6 sudo pvcreate /dev/sde
7 sudo pvcreate /dev/sdf
8 sudo vgcreate proVG /dev/sdd /dev/sde /dev/sdf
9 sudo lvcreate -L 5G -n proLV proVG
10 sudo mkfs.ext4 /dev/proVG/proLV
11 sudo mkdir /mnt/prodata
12 sudo mount /dev/proVG/proLV /mnt/prodata
13 echo "/dev/proVG/proLV /mnt/prodata ext4 defaults 0 0" | sudo tee -a /etc/fstab
14 sudo vgdisplay proVG
15 sudo lvextend -L 7G /dev/proVG/proLV
16 sudo resize2fs /dev/proVG/proLV
17 sudo yum install acl
18 sudo mount -o remount,acl /mnt/prodata
19 sudo useradd prouser
20 sudo passwd prouser
21 sudo setfacl -m u:prouser:rwX /mnt/prodata
22 getfacl /mnt/prodata
23 sudo setfacl -m o::--- /mnt/prodata
24 sudo lvextend -L 7G /dev/proVG/proLV
25 sudo resize2fs /dev/proVG/proLV
26 sudo pvs
27 sudo vgs
28 sudo lvs
29 df -h /mnt/prodata
30 getfacl /mnt/prodata
31 sudo lvdisplay /dev/proVG/proLV
32 history
```